

Autumn Science: Explain, Check, Improve

I can use real autumn science evidence to explain one linked idea, note a limit, and improve my work after feedback.



[Open the original interactive lesson](#)

Start with what is real

Get your W7A map. Put one marker by the link you trust most. Put one by a link that needs work.

Inherited Use authentic prior work, source cards or school-held records. Keep its date, page, filename or lesson title attached. Do not copy from memory when the genuine record is available.

Choose one strong link Point to the claim, then the evidence and the reason that joins them. Say whether the link is clear, partial or still uncertain. A precise symbol or label is enough.

Start with what is real

Choose one repair point Mark a missing locator, a claim that goes too far, a reason that is unclear or a limit that has disappeared. Name the problem without fixing it yet.

Prepare the comparison Keep the first version visible beside an empty change box. The first version is evidence of your thinking; it must not be erased when you improve it.

Supported

Use two prepared markers: STRONG LINK and CHECK THIS. Point to the exact words, number, label or model feature that made you choose.

Start with what is real

Standard

Write a short locator and one sentence explaining why the selected link is useful. Keep one honest uncertainty beside it.

What would count?

Picture first: the same four topics. Today: one linked idea, its evidence, and its limit.

Read two accounts. One repeats a fact. One names evidence and links it to the claim. Find the sentence that shows the science thinking.

Account A

“The model has arrows.” Decide what is present and what is missing. Does this locate evidence, explain a link or state only a visible feature?

Account B

What would count?

“The arrows point from the food to the animal, so the model shows the direction of energy transfer.” Identify the evidence words and the reasoning words.

- A. Account A gives enough evidence
- B. Account B shows the link

What would count?



Four autumn topics

Watch the decision trail

Limit finder

Choose a claim, then the honest limit that belongs beside it.

Watch the decision trail

- A. the globe-and-lamp model explains day and night
- B. the rover test compared two surfaces
- C. the classroom ball stood for Earth
- D. the climate figures we used were simplified
- E. a model is not to scale and simplifies the axis
- F. three repeats on two surfaces — a pattern, not a law
- G. a smooth ball hides mountains and slight flattening
- H. real climate data is noisier than a diagram

Watch the decision trail

Neutral model: a labelled food-chain card shows grass → rabbit → fox. The claim is narrow: the arrows show the direction of energy transfer in this model.

1 · State a claim that the record can answer. “In this model, energy is transferred from the grass to the rabbit and then to the fox.”

2 · Name the evidence: source, result, note or labelled model. “The labelled arrows on source card FC-2 point grass → rabbit → fox.” A source ID and the visible feature make the record locatable.

Watch the decision trail

3 · Use because or therefore to link them. Then add a fair limit. “The arrows support the transfer direction, but this card does not show the amount of energy transferred.” The limit prevents the evidence carrying a claim it cannot support.

Before feedback Keep the original claim, locator, reason and limit visible. Do not tidy it away while the model is being checked.

After feedback Change one precise part, then name what changed and why. The response is guidance, not a score or an assessment decision.

Choose, then justify

Choose the move that makes the science more true, not just longer.

Clear Add the missing locator, label or science word so another reader can find the record and follow the link.

Fair Narrow the claim, restore an uncertainty or state what the record cannot show.

Wider Connect the same reasoning to a second authentic record without inventing a result or changing the first one.

Checked change Keep the first version. Write the new version beside it and name the response you used.

Accuracy includes limits

Read or hear the response. It is not a score. Choose: make clear, make fair or make wider.

First version: "The food chain proves all foxes eat rabbits." Keep it visible even though it goes beyond the model.

Response: "Make fair: say what this card shows, not what every fox does." The response identifies the boundary; it does not write the learner's revision.

Checked revision: "This card shows a possible grass → rabbit → fox feeding link." Record the exact change. Keep the first draft so the change can be seen.

Accuracy includes limits

Keep

Do not add

Build the route together

1 · Locate

Find the genuine record, source or prior idea. Keep its date, filename, page, lesson title or source-card ID attached.

2 · Connect

Explain what it supports and how. Use a label, result, observation or model feature that another reader can find.

3 · Limit

Build the route together

Keep uncertainty, missing evidence and adult decisions visible. Say what the record cannot show.

4 · Receive Ask for one bounded response: make clear, make fair or make wider. The learner chooses whether to use it and may ask for it again.

5 · Revise Change one precise part without erasing the first version. Name the words, label, locator or limit that changed.

6 · Check Read from claim to evidence to reason to limit. Remove any invented detail and keep “not found yet” when a source is absent.

Build the route together

7 · Hand forward Name the genuine record, learner contribution, adult support, saved change and next authorised action.

Use the W7A evidence map, local GROW Science W8–W13 lessons and the pack's source cards.

Make the evidence trail usable

Make one evidence-led account from real autumn work. Include a claim, the place to find the evidence, the link, one limit and one clear change after a response. You may write, draw and label, use the school's approved audio route, or explain in private to an adult.

Use only Genuine prior work, local source cards, labelled models, real recorded results or a correction that is still visible.

You may Point, sort, draw, label, write, type, use the approved audio route or dictate your exact words.

Make the evidence trail usable

Keep separate Your choice and meaning; shared rehearsal; adult preparation, prompting, exact-word scribing and filing.

Supported

Use a claim/evidence/reason frame and one evidence item. Respond in private to one feedback choice.

Write or point one box at a time: My claim is ... My evidence is ... This supports it because ... It does not show ... The response was ... I changed ...

Standard

Make the evidence trail usable

Explain a linked idea with two evidence links, one limit and a saved change.

Add a locator for each evidence item. Compare what the two records show, then choose one clear, fair or wider revision.

Stretch

Judge two accounts or two pieces of evidence. Then use the same thinking for a new linked question.

Explain which evidence is more useful for the claim and why. Transfer the reasoning route, not an invented result, to the new question.

Make the evidence trail usable

Claim check Make it narrow enough for this record to answer. Replace “proves,” “always” or “all” when the evidence cannot carry those words.

Locator check Name the page, date, filename, lesson, result row, model label or source-card ID. If it is absent, write “not found yet.”

Link check Point to the exact evidence feature, then explain how it supports, refines or challenges the claim. Do not add a hidden detail.

Limit check Name what the record cannot show, what is uncertain or what another genuine record would need to check.

Make the evidence trail usable

Response check Record the bounded response as make clear, make fair or make wider. It is guidance, not a mark or award decision.

Revision check Keep the first version. Show the new version beside it and name the precise change that was made.

Checking questions: Can I find every source I named? Does each science claim match what the source can show? Have I used evidence words and reasoning words? Is the limit still visible? Can someone see the first version, the response and my change? Have I marked any shared decision or adult support honestly? If one answer is no, return to that box only. Keep the rest of the genuine trail unchanged.

Make the evidence trail usable

Use the honest route, not a shortcut.

One evidence link, one limit, one next step

Show the before and after change. Complete: 'My evidence supports... It does not show... I changed... because...'.

My evidence link Name the claim, genuine record and reason. Point to the exact source, result, label or model feature.

My honest limit Say what this evidence does not show or what still needs checking. Keep uncertainty visible.

My checked change Show the first and revised versions. Name the clear, fair or wider response you used and the precise change.

One evidence link, one limit, one next step

Adult response The receiving adult names back the learner's link, limit and saved change.
Adult preparation, prompting, scribing and filing stay separate.